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Alkanes – What Are They?

Alkanes are the hydrocarbon chains that make up the primary constituents of fuels. Learn more about what alkanes are



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Alkanes: The Building Blocks of Fuels

Fuels are mixtures of hydrocarbon chains of varying lengths that meet the formula C_nH_{2n+2} . Molecules that meet this formula are known as 'alkanes', and these hydrocarbons are found in all organic matter. They also make up crude oil, and when crude oil is refined through fractional distillation, these alkane mixtures can be separated into many useful products – such as **diesel**, **gas oil**, **kerosene**, **marine gas oil** and **lubricants** among others.



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are used in chemical processes, not fuels.

When a mixture of alkanes forms a distillate a flashpoint of higher than 42°C they are classified as fuel oils. [Learn more about fuel oil here.](#)

If you want to know more about the process of refining crude oil that allows alkanes to create useful fuels and oils, read our guide to [crude oil fractional distillation](#).

Table of alkane hydrocarbons

The below lists the first 20 alkanes, although there are many more. Each type of fuel will have mixtures of different alkane chains, and these mixtures will give the fuel its desired properties.

Alkane	Molecular Formula	Composition	Carbon atoms in chain	Hydrogen atoms in chain
Methane	CH ₄	CH ₄	1	4
Ethane	C ₂ H ₆	CH ₃ -CH ₃	2	6
Propane	C ₃ H ₈	CH ₃ -CH ₂ -CH ₃	3	8
Butane	C ₄ H ₁₀	CH ₃ -2(CH ₂)-CH ₃	4	10
Pentane	C ₅ H ₁₂	CH ₃ -3(CH ₂)-CH ₃	5	12
Hexane	C ₆ H ₁₄	CH ₃ -4(CH ₂)-CH ₃	6	14
Heptane	C ₇ H ₁₆	CH ₃ -5(CH ₂)-CH ₃	7	16
Octane	C ₈ H ₁₈	CH ₃ -6(CH ₂)-CH ₃	8	18
Nonane	C ₉ H ₂₀	CH ₃ -7(CH ₂)-CH ₃	9	20

Alkane MENU	Molecular Formula	Composition	Carbon atoms in chain	Hydrogen atoms in chain
Decane	$C_{10}H_{22}$	$CH_3-8(CH_2)-CH_3$	10	22
Undecane	$C_{11}H_{24}$	$CH_3-9(CH_2)-CH_3$	11	24
Dodecane	$C_{12}H_{26}$	$CH_3-10(CH_2)-CH_3$	12	26
Tridecane	$C_{13}H_{28}$	$CH_3-11(CH_2)-CH_3$	13	28
Tetradecane	$C_{14}H_{30}$	$CH_3-12(CH_2)-CH_3$	14	30
Pentadecane	$C_{15}H_{32}$	$CH_3-13(CH_2)-CH_3$	15	32
Cetane (Hexadecatane)	$C_{16}H_{34}$	$CH_3-14(CH_2)-CH_3$	16	34
Heptadecane	$C_{17}H_{36}$	$CH_3-15(CH_2)-CH_3$	17	36
Octadecane	$C_{18}H_{38}$	$CH_3-16(CH_2)-CH_3$	18	38
Nonadecane	$C_{19}H_{40}$	$CH_3-17(CH_2)-CH_3$	19	40
Eicosane	$C_{20}H_{42}$	$CH_3-18(CH_2)-CH_3$	20	42

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